Deepak Mallampati

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PROFESSIONAL SUMMARY

LLM deployment engineer (MS CS, 2.5+ years) building production RAG systems, agentic LLM workflows, and evaluation pipelines for enterprise environments. Delivered ~35% retrieval precision gain, >30% hallucination reduction, and 25% agent response stability improvement. Deep expertise in LangChain, LlamaIndex, vector search, structured outputs, and prompt engineering. Expert at instrumenting, evaluating, and shipping AI systems into regulated production - bridging RL-Ops platform capabilities and real customer deployments.

CORE COMPETENCIES

RAG Systems • Agentic LLM Workflows • LLM Evaluation Pipelines • LangChain / LlamaIndex • Vector Search & Embeddings • Prompt Engineering • LLM Fine-tuning (PyTorch) • FastAPI / Flask / Docker • Python • Multi-agent Orchestration • Production AI Deployment • Structured Outputs / Guardrails

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting

- Built **production RAG systems** for clinical knowledge retrieval with recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by **~35%**, reduced irrelevant retrieval by **>30%**; retrieval-grounded response alignment exceeded **90%** in evaluation.
- Developed **agentic LLM workflows** with structured reasoning, tool discipline, and grounding rules for multi-step healthcare queries; improved agent response stability by **~25%** and reduced hallucinations by **>30%**.
- Built **LLM evaluation pipelines** with precision/recall tracking, hallucination measurement, grounding alignment scoring, and stakeholder-facing audit trails.
- Packaged LLM inference as **FastAPI/Flask RESTful services** with Docker containerization, structured logging, and load simulation; reduced post-deployment defects by **~30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage, evaluation audit trails.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer - Applied NLP, Video Analytics

- Built **transformer-based domain adaptation** for video summarization (5,000+ sessions); hierarchical summarization and timestamp-aligned extraction; **60-70% reduction** in review time, **~85% alignment** with human-curated highlights.
- Implemented **RAG-grounded document Q&A**; reduced hallucinated outputs **~30%**, raised contextual accuracy **>85%**.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern)

- Optimized SQL/T-SQL stored procedures; reduced query execution time **~20%**.
- Built Jenkins CI/CD pipelines for schema deployments; reduced deployment errors **>30%**.

KEY PROJECTS

Agentic Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN-based duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Manga Lens - Multi-Provider AI Inference Chrome Extension

Production Chrome extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider payload handler - WebP for cloud, JPEG for Ollama to avoid CUDA crash. Seven-day translation cache, per-domain configs for 29 sites. Demonstrates cross-model inference architecture and provider abstraction.

Dream Decoder - Multimodal AI with Perplexity Sonar

FastAPI + React/TypeScript app using Perplexity Sonar API for semantic retrieval + Replicate for image synthesis. Introduced **intermediate structured prompt transformation layers**; improved contextual alignment **~30%** and first-pass image success **~25-30%** over naïve direct-prompt orchestration.

Driver Drowsiness Detection - YOLOv8 Real-Time System

Replaced two-stage CNN pipeline with **unified YOLOv8**; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives by **~25%**.

EDUCATION

Master of Science, Computer Science - *Kent State University, Ohio* *Aug 2023 - May 2025*

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL, TypeScript, React

LLM & GenAI: RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering, evaluation pipelines, structured outputs, guardrails, vector search, embeddings

ML: PyTorch, scikit-learn, XGBoost, HuggingFace Transformers, fine-tuning

Infra: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, cloud-ready deployment

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required